

Use-Case Flow Specification

Problem Statement #45 – Multi-Cloud Asset Cost Optimization Portal

NAME: R Vinay SRN: PES1UG24CS353 SECTION: F

Use Case: UC-06 – Resolve Flagged Asset

Primary Actor: Cloud Administrator

Secondary Actors: FinOps Analyst (notified of resolution), Cloud Billing Provider (executes termination/deallocation)

Preconditions

- Billing data has been ingested and categorized (UC-01, UC-02).
- At least one compute instance or storage volume has been flagged as idle or unattached (UC-03).
- The Cloud Administrator is logged in with permission to act on the relevant cloud account/project.

Postconditions

- Success: The flagged asset's status is 'Resolved', the underlying cloud resource has been terminated/deallocated, and the achieved savings are recorded.
- Alternate: The flagged asset's status is 'Excluded' as a false positive, no cloud resource is changed, and the exclusion is logged for audit.

Main Success Scenario

1. Cloud Administrator opens the list of flagged idle/unattached assets in the portal.
2. System displays asset details: type, region, estimated monthly savings, and days idle/unattached.
3. Cloud Administrator selects a flagged asset and reviews the recommendation.
4. Cloud Administrator selects “Terminate/Deallocate Asset”.
5. System validates that the administrator has permission to act on that cloud account/project.
6. System sends a termination/deallocation request to the corresponding cloud provider API (AWS/Azure/GCP).
7. System updates the asset status to 'Resolved' and records the estimated cost savings achieved.
8. System notifies the FinOps Analyst that the flagged asset has been resolved.
9. Use case ends successfully.

Alternate Flow – A1: Administrator Marks Asset as False Positive

- A1.1 (branches after Step 3): Instead of terminating, Cloud Administrator selects “Mark as False Positive” and enters a required justification comment.
- A1.2: System records the false-positive decision and excludes the asset from idle/unattached scans for a 30-day cooldown period.
- A1.3: System updates the asset status to 'Excluded' and logs the justification for audit.
- A1.4: Use case ends without any change to the underlying cloud resource.